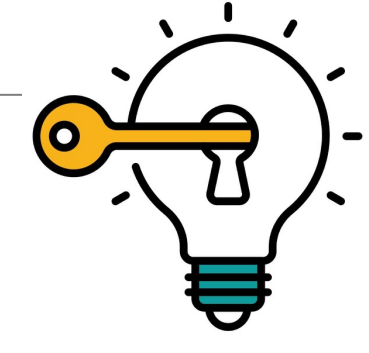


CoP: Coordinated Perturbation for Controlled Disclosure Under Local Differential Privacy

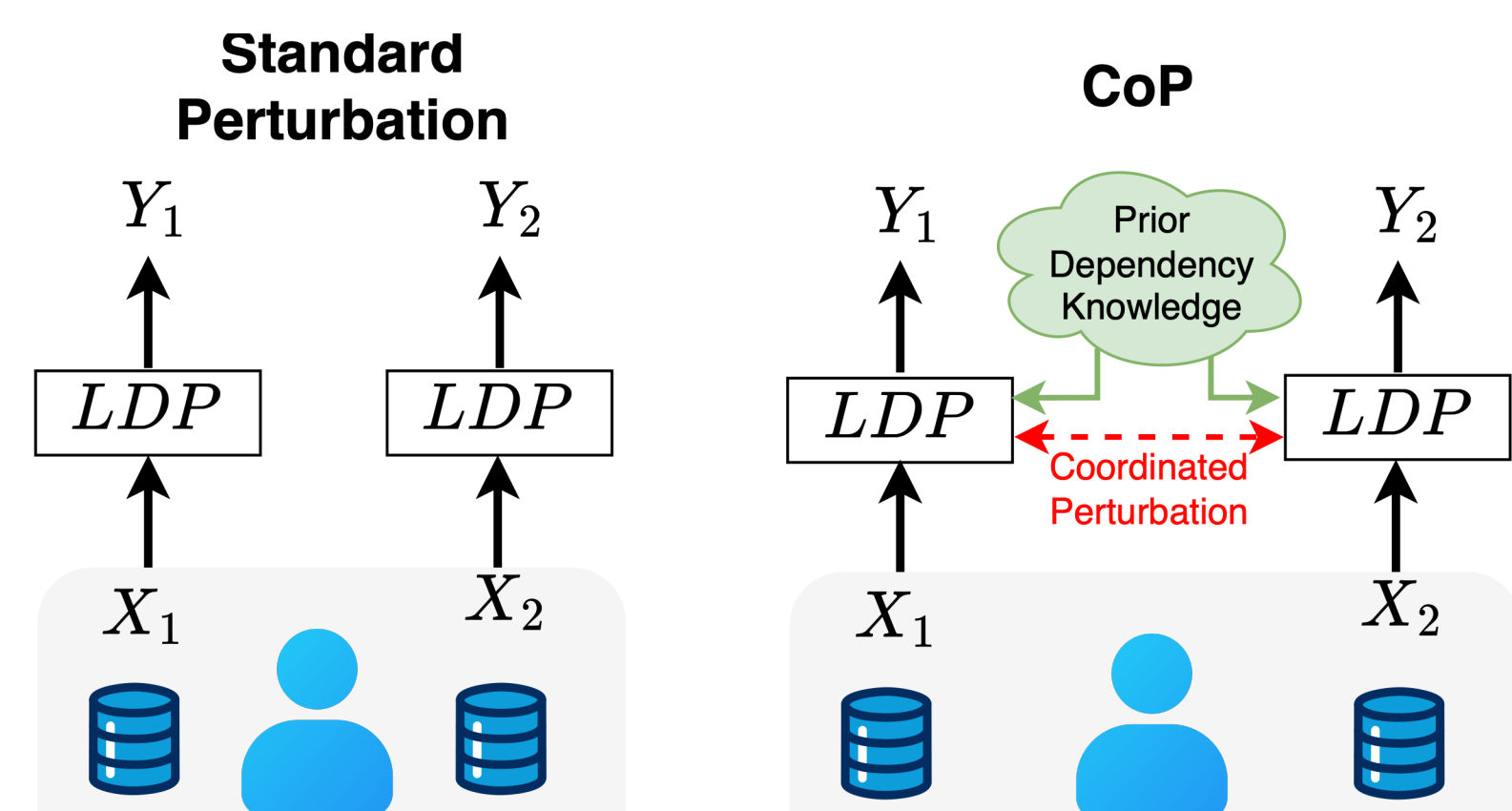
Sandaru Jayawardana (The University of Sydney, Australia), Ming Ding (CSIRO, Australia), **Kanchana Thilakarathna** (The University of Sydney, Australia)

Key Takeaway

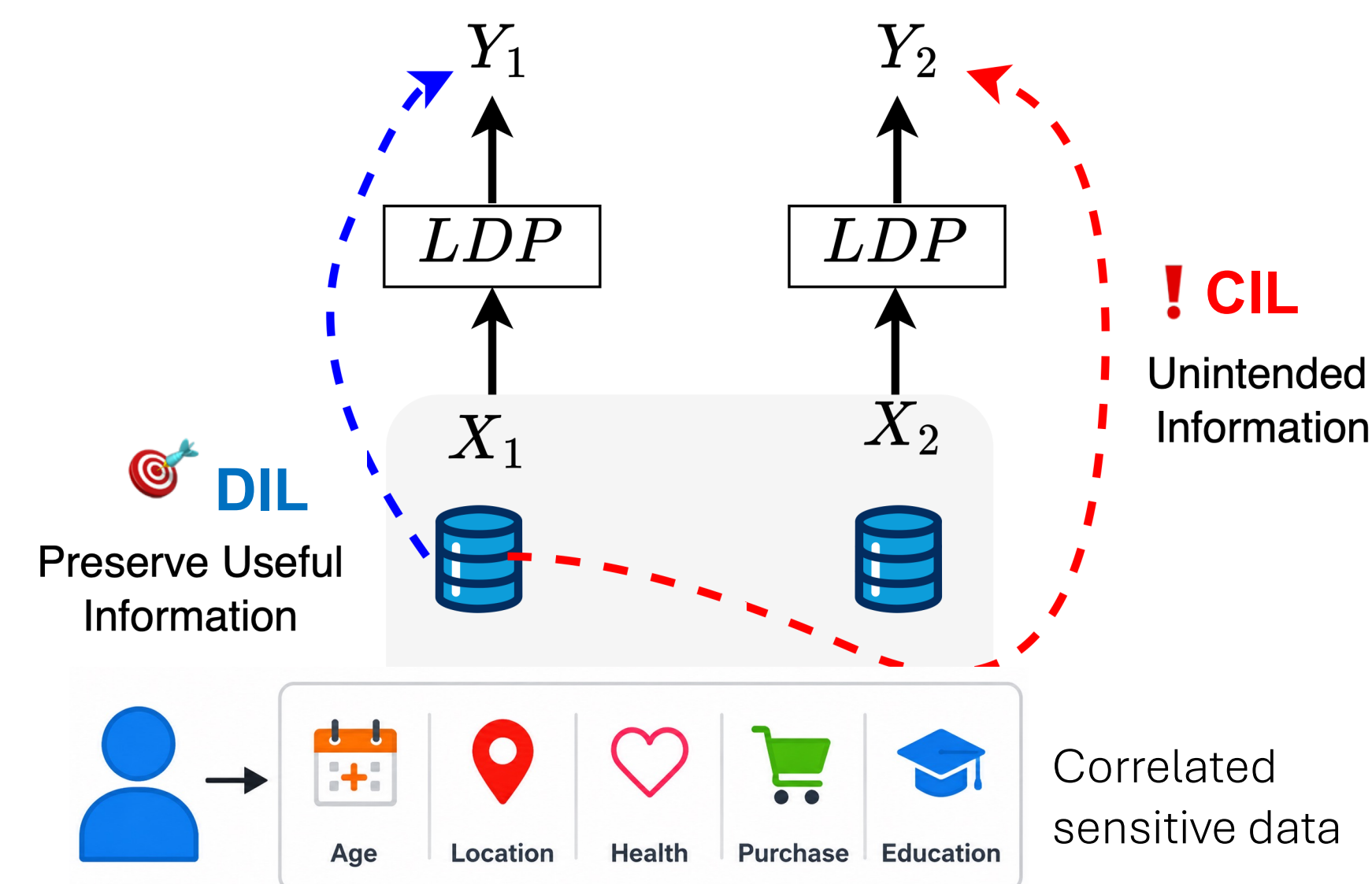


Attribute dependencies can cause privacy leakage even under Local Differential Privacy (LDP).

CoP uses prior dependency knowledge to coordinate perturbations, reducing CIL while preserving utility.

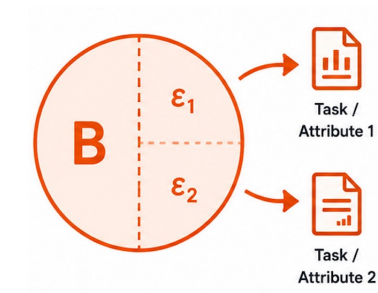


1 The Problem

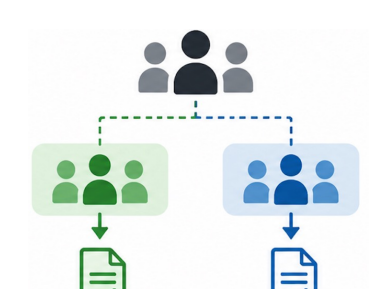


Correlations in the underlying data allow an attacker to infer private info.

2 Related Work



Budget splitting reduces leakage but increases noise and can lower utility.



Population splitting (i.e., random sampling) reduces per-user exposure but may lose valid unit-level multidimensional records.

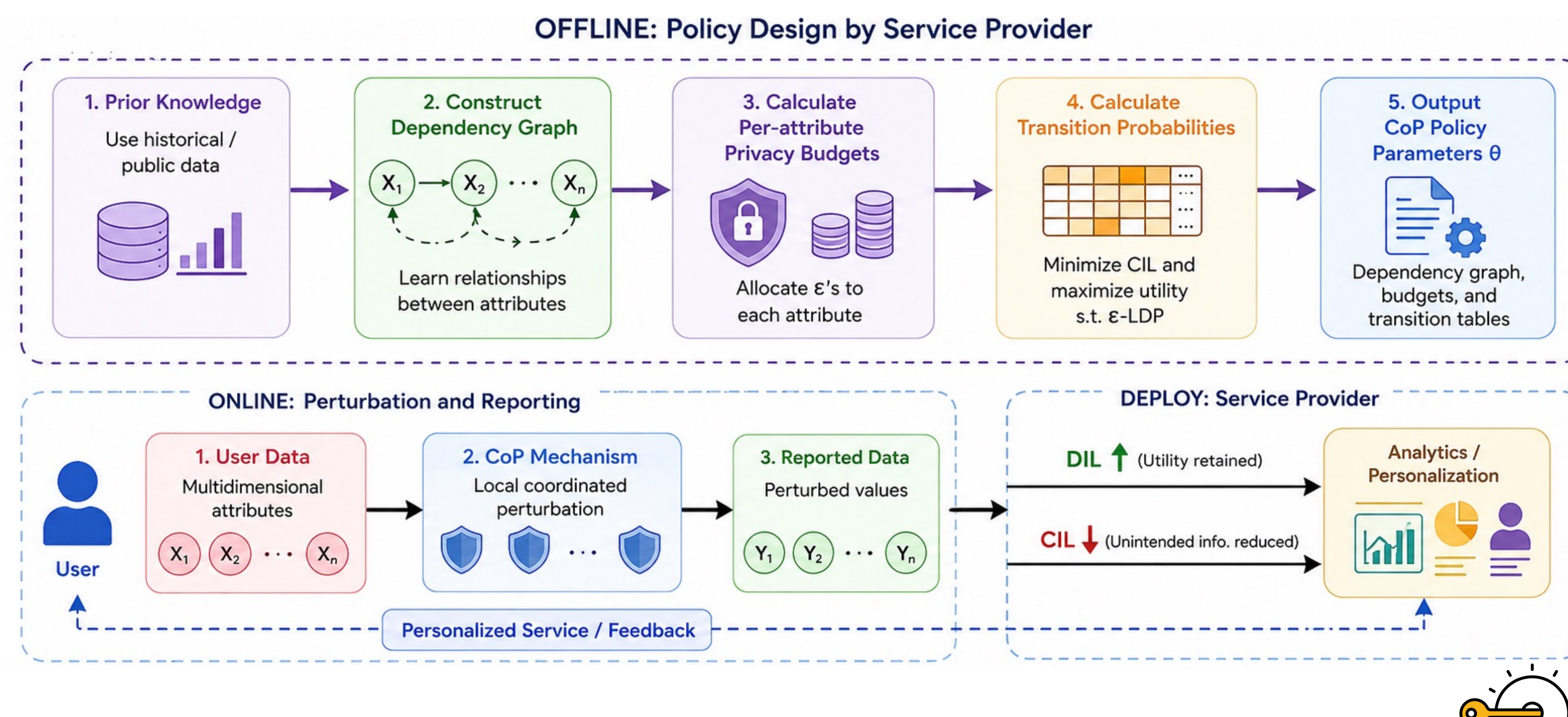


Joint perturbation captures attributes together but overlooks correlation-induced leakage.

Main Contributions

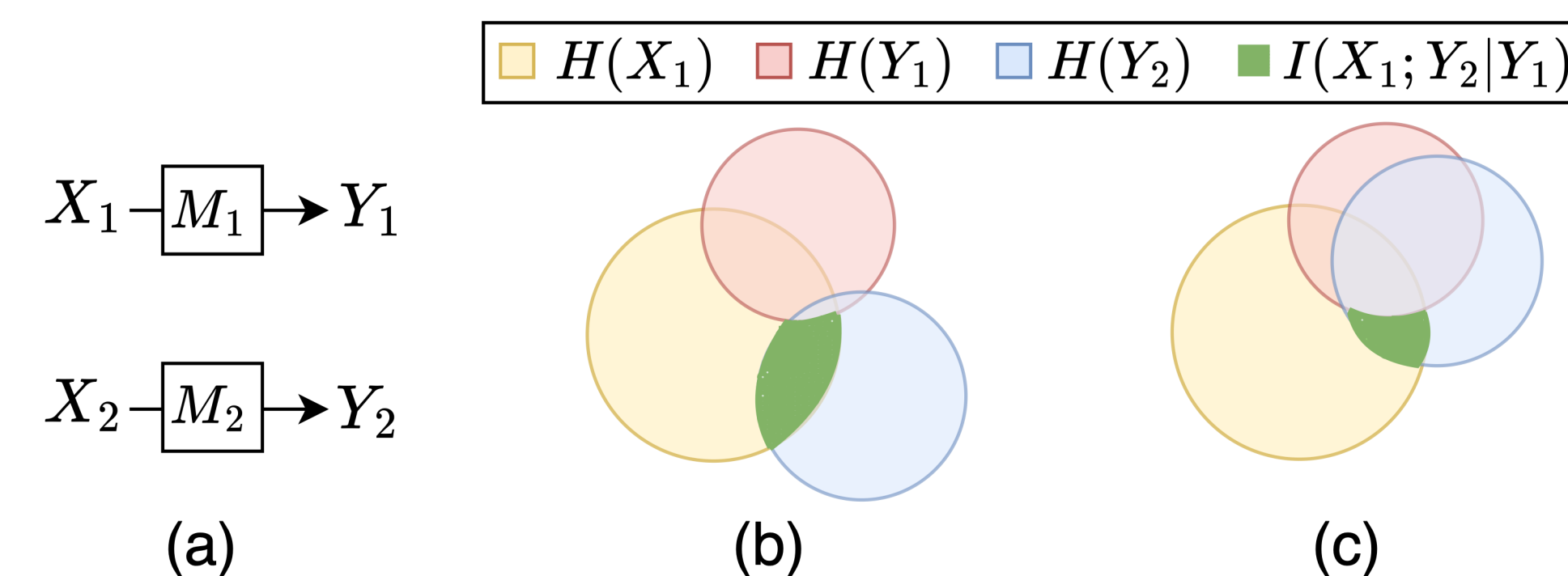
- Decompose LDP leakage into utility-relevant **Direct Information Leakage (DIL)** and unintended **Correlation-induced Information Leakage (CIL)**
- Propose CoP, a prior-knowledge-aware coordinated perturbation mechanism to reduce CIL while preserving DIL
- Prove CoP's ϵ -LDP guarantee and show that it improves the **privacy-utility trade-off** and reduces attribute-inference attack success

3 CoP Overview



4 CoP Core Intuition

Independent perturbation can make correlated reports act as multiple independent sources of evidence about a hidden attribute.



Entropy Venn diagram for $H(X_1)$, $H(Y_1)$, and $H(Y_2)$

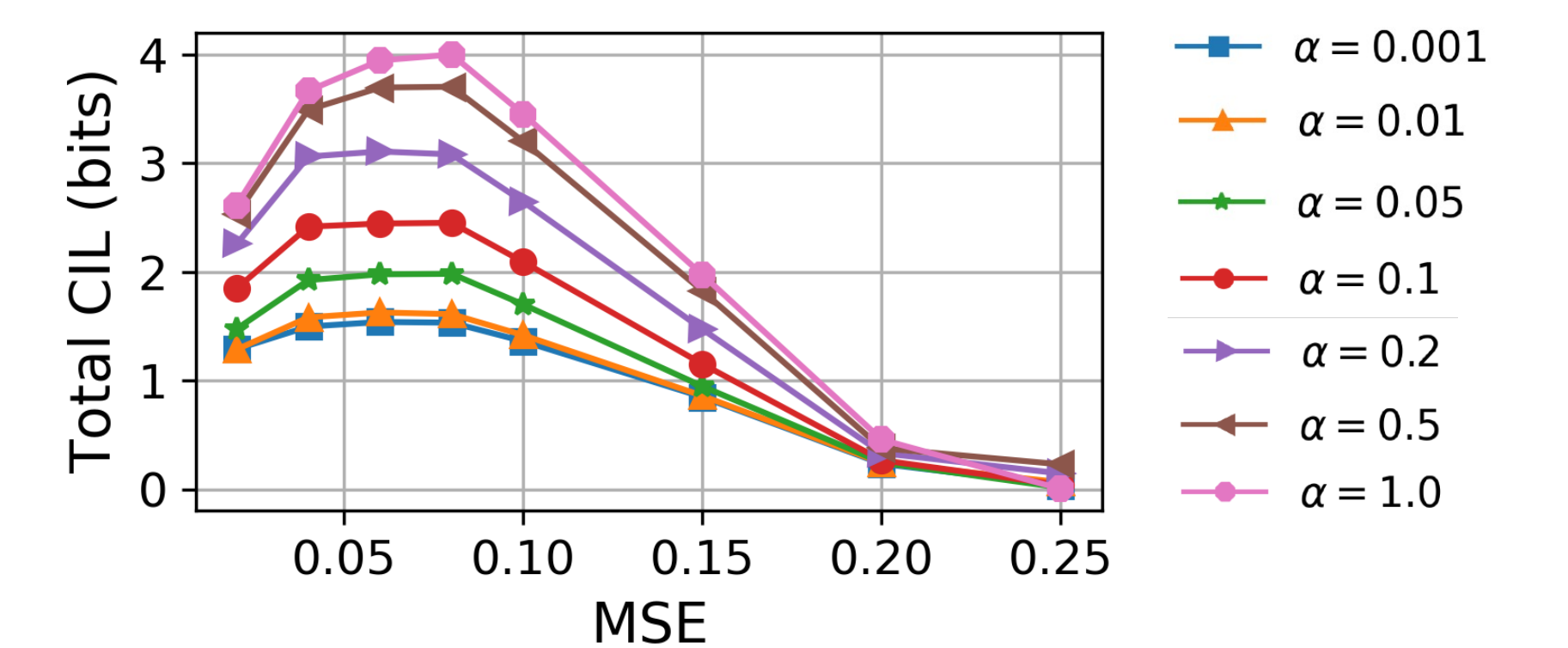
Illustrative Example

- Two attribute perturbations using LDP mechanisms.
- "Green region" illustrates the $CIL(X_1)$, $I(X_1; Y_2 | Y_1)$ (before increasing $I(Y_1; Y_2)$).
- Increased $I(Y_1; Y_2)$ by coordinated perturbation. The intersected area between $H(Y_1)$ and $H(Y_2)$ is larger than Fig. (b).

"Green region" $CIL(X)$ reduces as $I(Y_1; Y_2)$ increases

5 Results

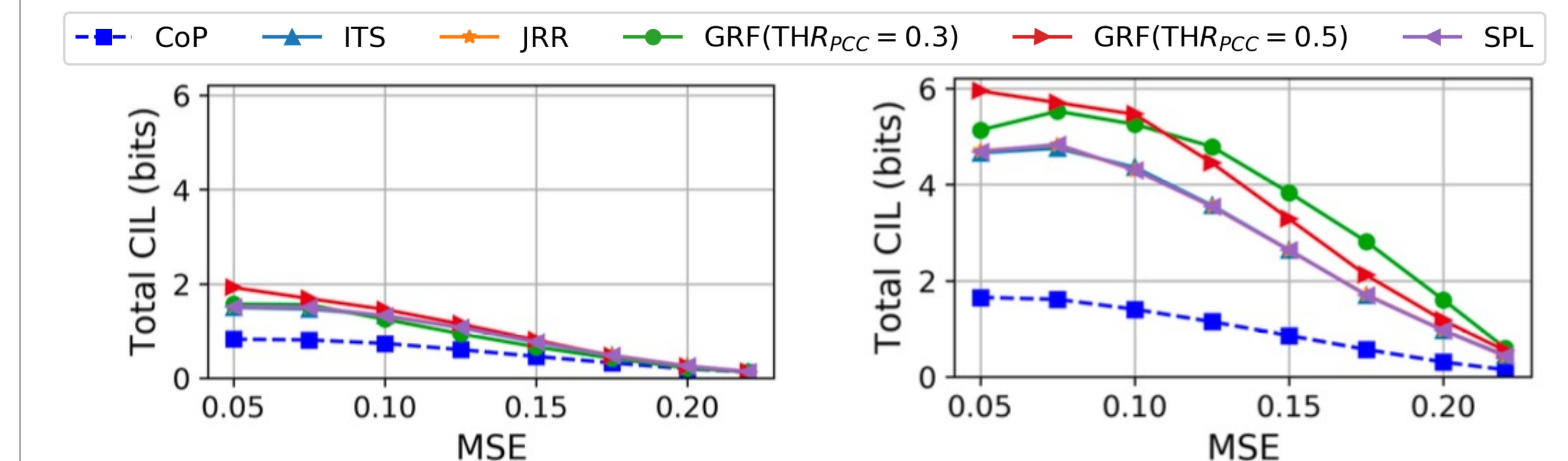
5.1 Effect of Dependency Calibration



The strength of coordination among attributes decreases as α increases

Calibrated coordination between perturbed outputs can reduce CIL.

5.2 Effect of Correlation Strength

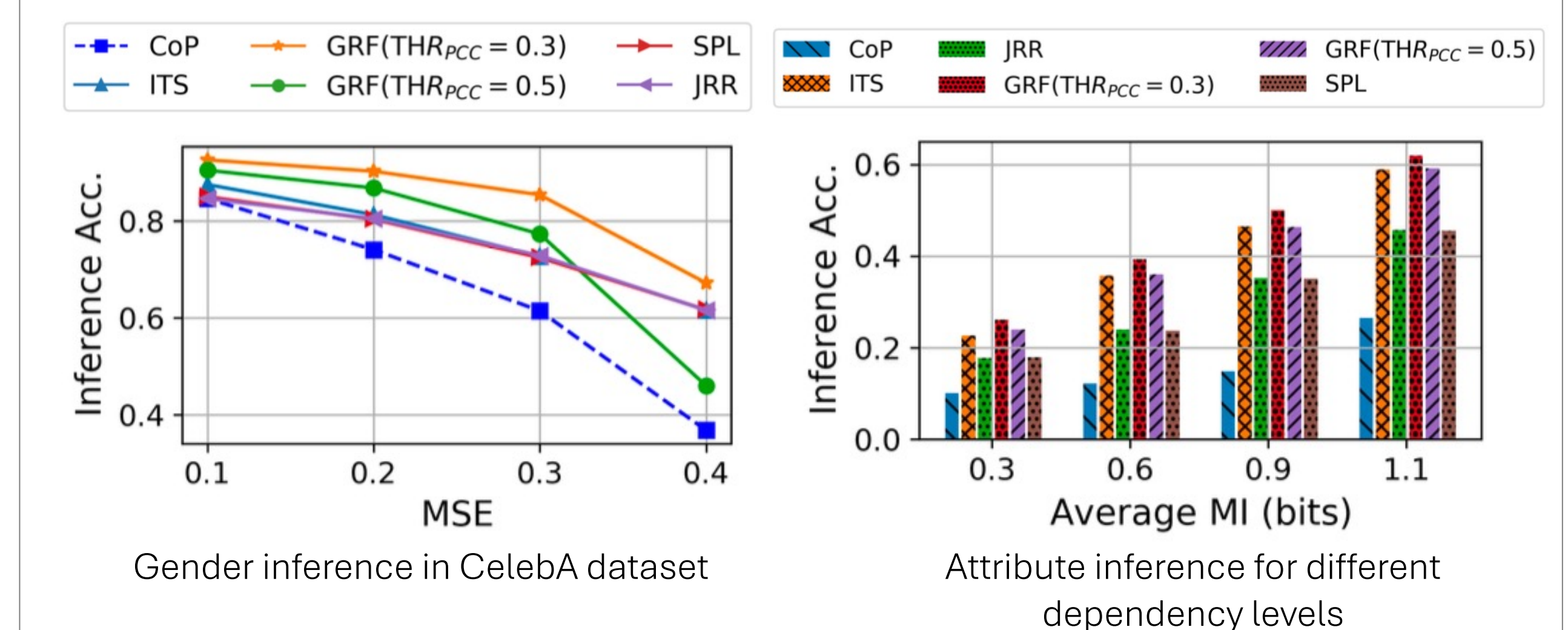


(a) Avg. MI = 0.3 bits

(d) Avg. MI = 1.1 bits

Stronger correlations enable more effective coordinated perturbation.

5.2 Attribute Inference Attacks with CoP



CoP lowers attribute inference providing stronger protection against correlation-based inference.

Further Information

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